

AYUSH UPADHYAY

Python Backend Developer | FastAPI, Django, ReactJS, Microservices & Generative AI

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SUMMARY

Python Backend Developer with hands-on experience across the full software development life cycle, specializing in scalable API development with FastAPI and Django and the design of high-performance, microservice-based backend architectures. Comfortable working with REST and exploring gRPC/GraphQL for end-to-end API integration, with growing front-end exposure in ReactJS for connecting UI to backend services. Strong experience with PostgreSQL, MySQL, and MongoDB for database modeling, query optimization, and caching (Redis). Skilled in building secure authentication systems using JWT and Role-Based Access Control (RBAC), centralized exception handling, and OpenAPI documentation. Hands-on with Docker, Linux production deployments, and VPS hosting. Currently expanding expertise in Generative AI and LLM-powered systems, working with OpenAI APIs, LangChain, Hugging Face, and Ollama to build intelligent backend solutions that combine traditional engineering with AI capabilities. Strong problem-solving, communication, and organizational skills honed through cross-functional collaboration.

TECHNICAL SKILLS

Programming Languages	Python, C, C++, JavaScript
Backend Frameworks	FastAPI, Django, Flask
Frontend Frameworks	ReactJS, AngularJS (familiar), HTML, CSS, Bootstrap
API Protocols	REST, gRPC (familiar), GraphQL (familiar)
Machine Learning Tools	LangChain, Prompt Engineering
AI Platforms	OpenAI, Hugging Face, Ollama
Databases	PostgreSQL, MySQL, Oracle, MongoDB, Redis, ClickHouse, RabbitMQ
Big Data	PySpark
DevOps & Deployment	Docker, Git, Linux server deployment, VPS hosting

PROFESSIONAL EXPERIENCE

Junior Backend Developer — *Drona Foundations, Ahmedabad*

Jun 2025 – Jun 2026

ERP System — *Microservices & API Platform*

- Designed RESTful APIs and structured backend services with clean, microservice-based architecture, improving modularity, maintainability, and scalability of the ERP platform.
- Explored gRPC alongside REST for efficient inter-service communication between ERP microservices, evaluating trade-offs for internal service-to-service calls.
- Developed and maintained scalable microservices for the ERP platform using FastAPI and PostgreSQL, optimizing CRUD operations and enhancing database query performance.
- Implemented JWT-based authentication and Role-Based Access Control (RBAC) to secure APIs and manage user permissions across different ERP modules.
- Built centralized exception handling and request-validation layers, enhancing API reliability and ensuring consistent, well-structured error responses.
- Containerized microservices using Docker and deployed them on Linux production servers, enabling consistent environments and smoother release cycles.
- Created and maintained Swagger (OpenAPI) documentation for all APIs, improving end-to-end UI/API integration and collaboration between frontend and backend teams.

Python Developer — *Growthify Services, Indore*

Mar 2024 – Jun 2025

- Developed backend APIs using Django and FastAPI for internal web applications, with REST endpoints supporting frontend integration and external service consumers.
- Designed database schemas and wrote complex queries on MySQL and PostgreSQL, focusing on relational modeling and query performance.
- Implemented JWT authentication to secure user access across multiple internal tools.
- Contributed to projects including chat systems and an employee management tool, expanding their functionality and reliability.
- Collaborated using Git and agile methodologies, participating in code reviews and iterative feature delivery across the software development life cycle.

SELECTED PROJECTS

Real-Time Chat Application —

Stack: *FastAPI, WebSockets, MongoDB, JWT Authentication, REST APIs*

- Developed a real-time chat application using FastAPI, exposing both REST and WebSocket routes for messaging and room management.
- Designed MongoDB collections to store users, rooms, and messages with efficient indexing for fast retrieval.
- Implemented JWT-based authentication for secure user access and protected WebSocket connections.
- Built room-based messaging logic with connection management and message persistence.

Office Employees Management System (OEMS) —

Stack: *Django, Django ORM, MySQL, HTML, CSS, Bootstrap, ReactJS (UI enhancements)*

- Developed a web-based employee management system from scratch using Django, covering the full employee lifecycle.
- Implemented CRUD operations using Django ORM and QuerySet for managing employee records and related entities.
- Integrated authentication and authorization for secure, role-aware user access to administrative features.
- Designed responsive UI with HTML, CSS, Bootstrap, and ReactJS components, ensuring usability across desktop and mobile devices.
- Modeled and managed data in MySQL with normalized schemas for efficient storage and retrieval.

EDUCATION

Bachelor of Computer Applications (BCA)

2022 – 2025

Shri Vaishnav Vidyapeeth Vishwavidyalaya, Indore

CERTIFICATIONS

- Full Stack Web Development — Python
- Databases — NPTEL
- Full Stack Generative & Agentic AI with Python — Udemy